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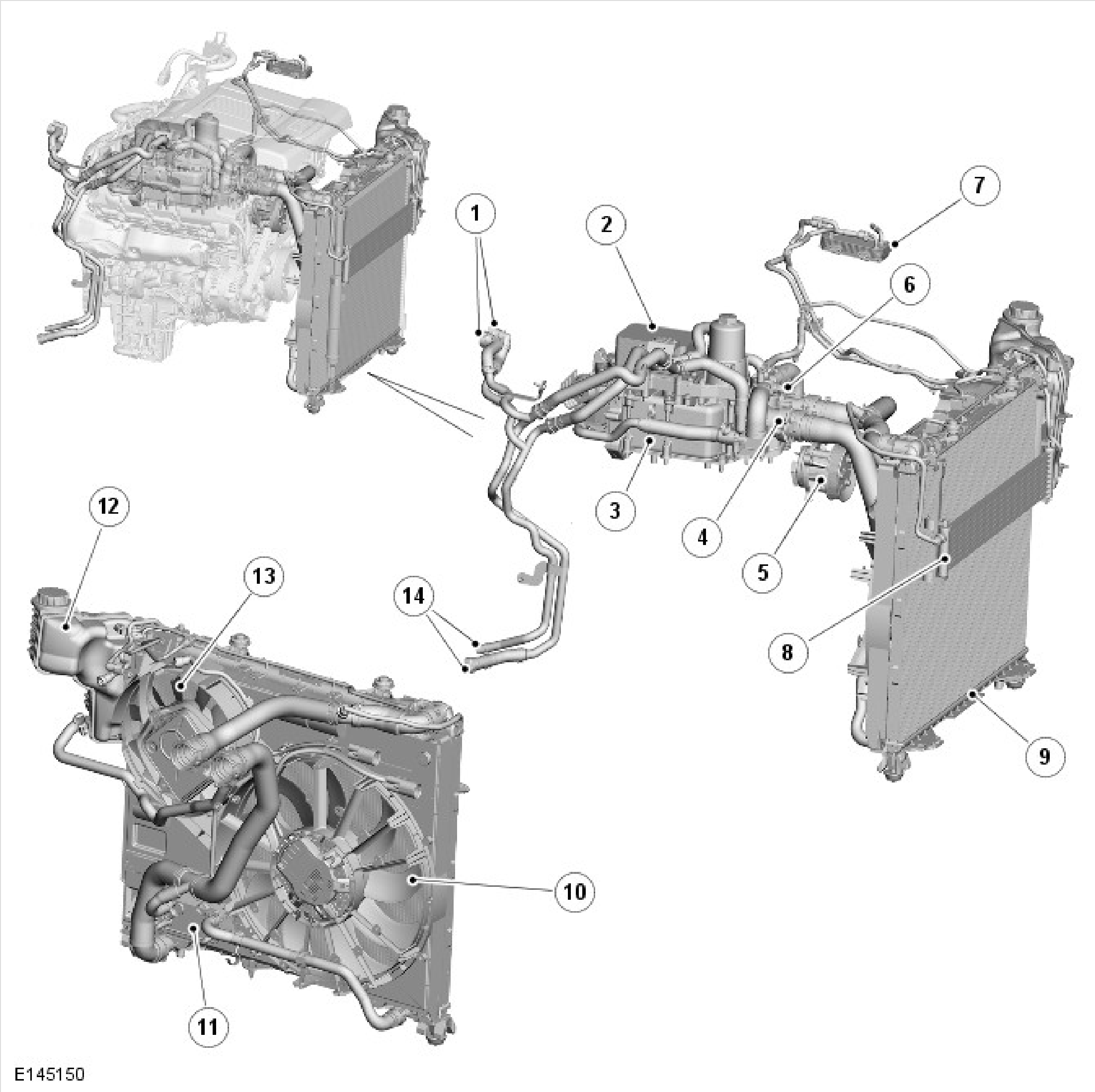
2015.0 RANGE ROVER (LG), 303-03

ENGINE COOLING - TDV8 4.4L DIESEL

COMPONENT LOCATION



ITEM	DESCRIPTION
1	Heater connections
2	Engine oil cooler
3	Exhaust gas recirculation cooler
4	Thermostat
5	Coolant pump
6	Engine coolant temperature sensor

7	Fuel cooler
8	Auxiliary fuel cooler
9	Radiator
10	Cooling fan 1
11	Transmission oil cooler
12	Expansion tank
13	Cooling fan 2
14	Auxiliary heater connections

OVERVIEW

The pressurized cooling system circulates coolant around the engine and interior heater circuits while the thermostat main valve is closed. The primary function of the cooling system is to maintain engine metal temperatures within an optimum temperature range under changing ambient and engine operating conditions. Secondary functions are to provide:

- Optimized cabin heating during the engine warm-up phase. For additional information, refer to:
  Heating and Ventilation (412-01A Climate Control, Description and Operation), Auxiliary Climate Control (412-02A Auxiliary Climate Control, Description and Operation).
- Automatic transmission fluid temperature control.
  For additional information, refer to: Transmission Cooling (307-02A Transmission /Transaxle Cooling - Vehicles With: 8HP45 8-Speed Automatic Transmission AWD, Description and Operation).
- Low temperature fuel cooling.
  For additional information, refer to: Fuel Tank and Lines (310-01D Fuel Tank and Lines - TDV8 4.4L Diesel, Description and Operation).
- EGR (exhaust gas recirculation) cooling.
  For additional information, refer to: Engine Emission Control (303-08E Engine Emission Control - TDV8 4.4L Diesel, Description and Operation).

During the initial engine warm-up, when the main thermostat is closed, warm coolant is directed to the ancillary heat exchangers:

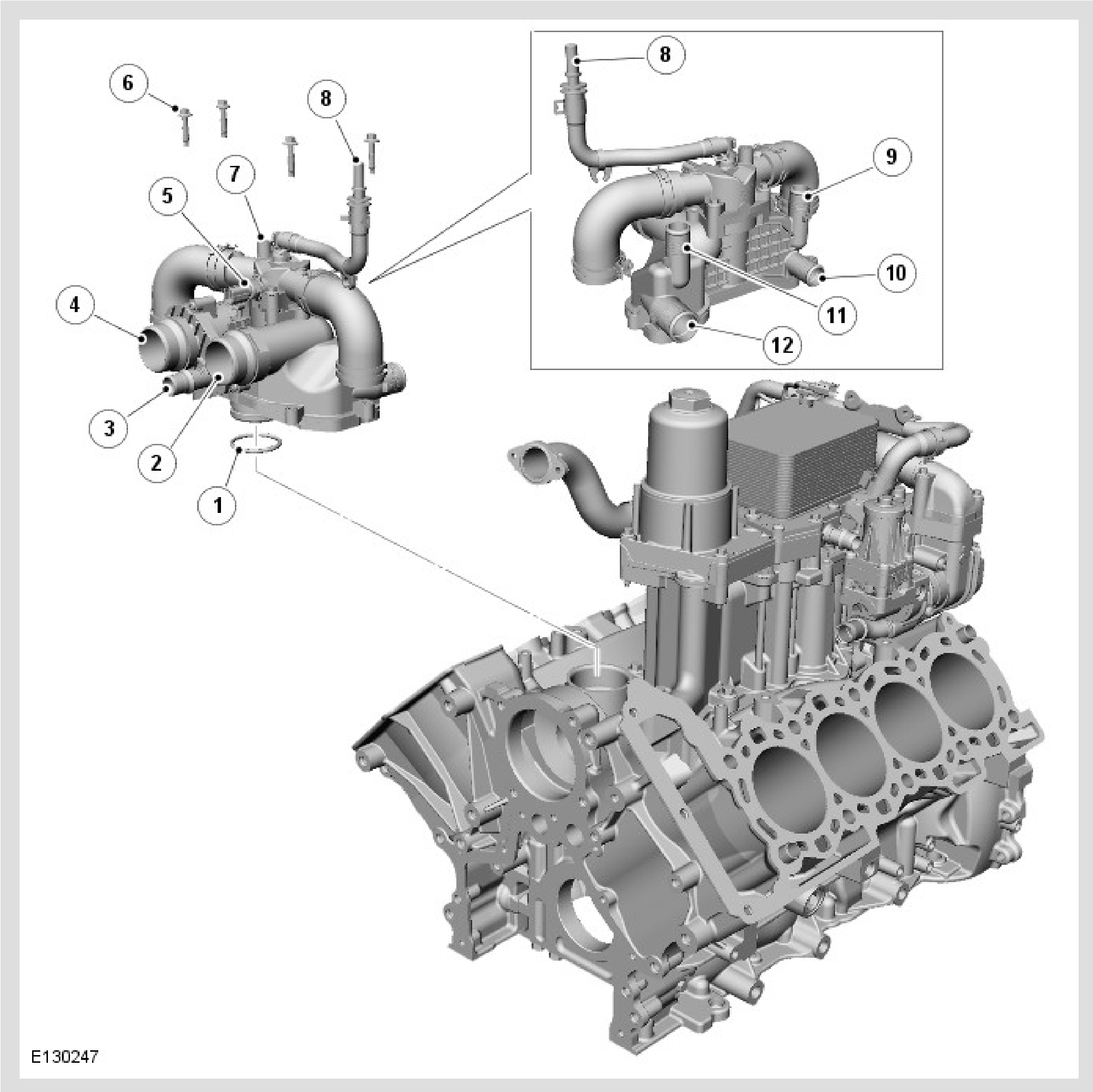
- Cabin heater
- Engine oil heat exchanger.

The main engine thermostat is of a pressure relief design. The prime function of this thermostat is to use the heater and ancillary circuit as the only radiator by-pass while the thermostat is closed.

The pressure relief thermostat gradually opens the main radiator by-pass circuit at engine speeds above 1500 rev/min. This system ensures that the maximum available coolant flow is directed to the heater and ancillary heat exchangers during warm-up without affecting the durability of the engine. By using this type of thermostat there is enough heater flow for all conditions without the need for a dedicated electric water pump.

DESCRIPTION

THERMOSTAT HOUSING



ITEM	DESCRIPTION
1	Seal
2	Connection for radiator upper hose
3	Connection for expansion tank hose
4	Thermostat/Connection for radiator lower hose
5	Engine coolant temperature sensor
6	Bolt (4 off)
7	Housing

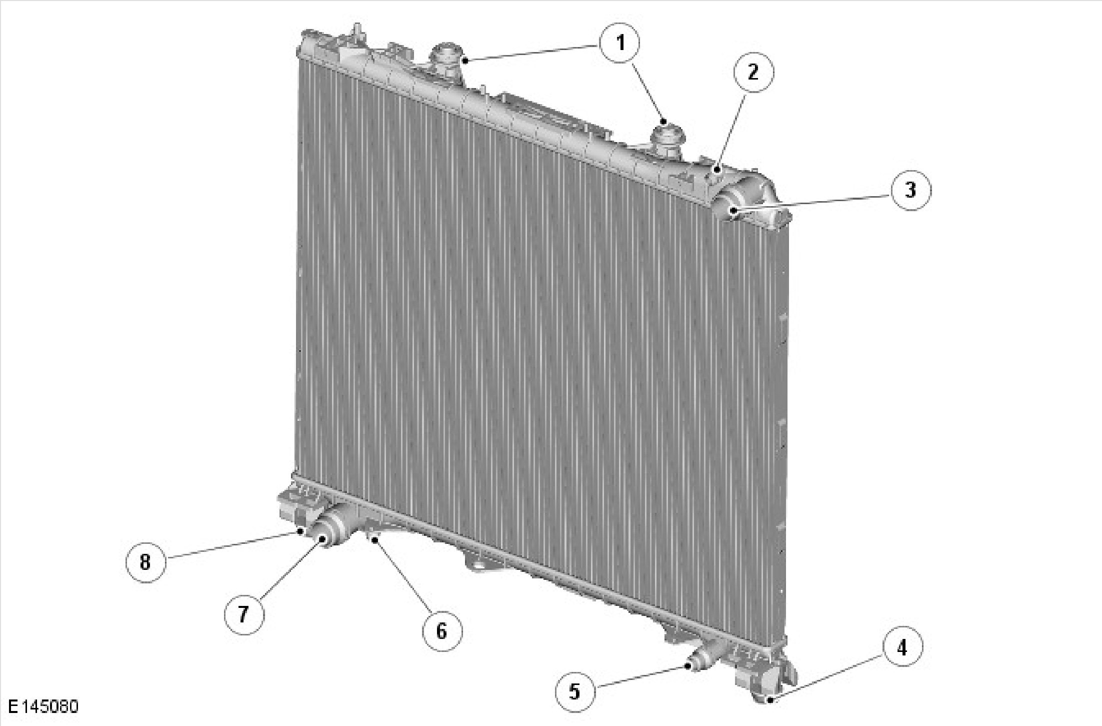
8	Coolant bleed to expansion tank
9	Connection for heater supply hose
10	Connection for EGR cooler supply hose
11	Connection for heater return hose
12	Connection for EGR cooler and engine oil cooler return hose

The thermostat housing contains the thermostat and connects the cooling channels of the cylinder block to the cooling channels of the cylinder heads. It also provides connections for:

- The radiator upper and lower hoses
- The expansion tank supply hose
- The EGR cooler supply hose
- The EGR cooler and engine oil cooler return hose
- The heater supply and return hoses
- A bleed hose.

An ECT (engine coolant temperature) sensor is located in the front of the thermostat housing. The ECT sensor is connected to the ECM (engine control module) which uses the coolant temperature information for a number of functions.
For additional information, refer to: Electronic Engine Controls (303-14E Electronic Engine Controls - TDV8 4.4L Diesel, Description and Operation).

RADIATOR

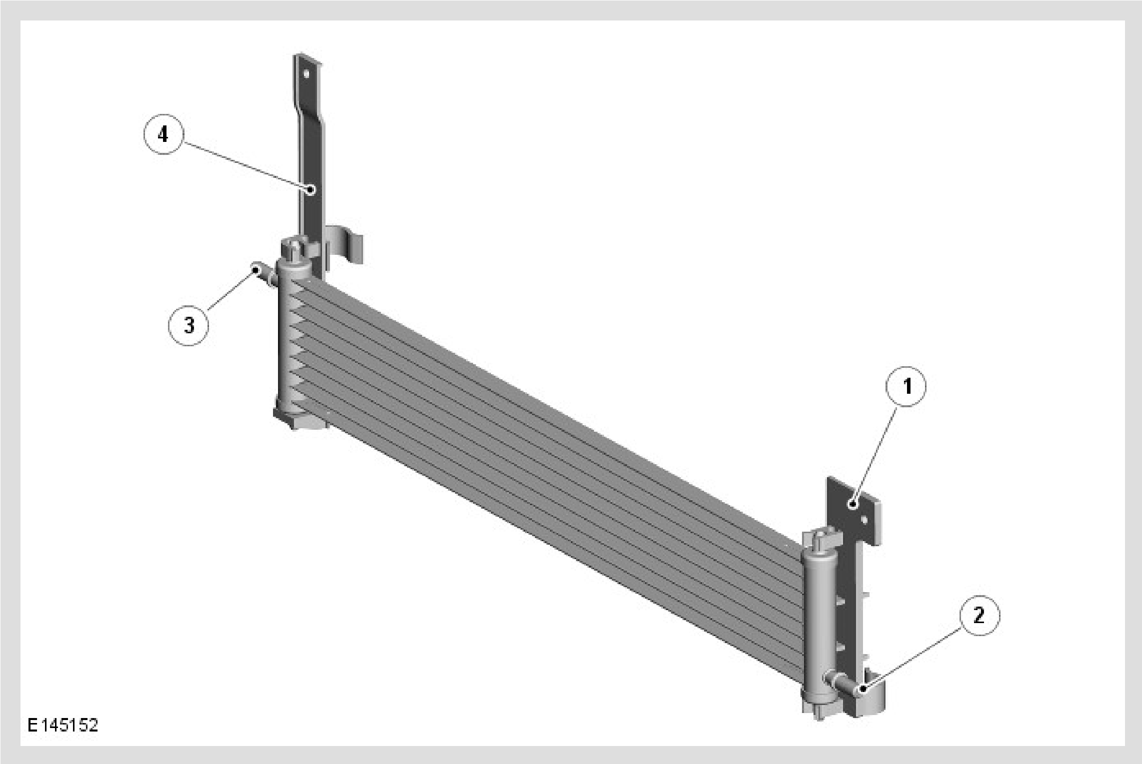


ITEM	DESCRIPTION
1	Upper mounts
2	Vent hose connection
3	Upper hose connection
4	Lower mount
5	Transmission hose connection
6	Drain screw
7	Lower hose connection
8	Lower mount

The radiator is a vertical-flow type with an integral transmission sub-cooler. The radiator is located in the vehicle by bushes attached to the end tanks. The lower bushes are installed in the front subframe. The upper bushes are installed in the front end carrier.

Upper and lower hose connections are incorporated into the upper and lower end tanks respectively. The lower end tank also incorporates a drain screw, and a connection for the coolant feed hose of the transmission fluid cooler. The upper end tank incorporates a connection for a vent hose to the expansion tank. The end tanks also incorporate mounting points for the cooling fan shroud and the A/C (air conditioning) condenser.

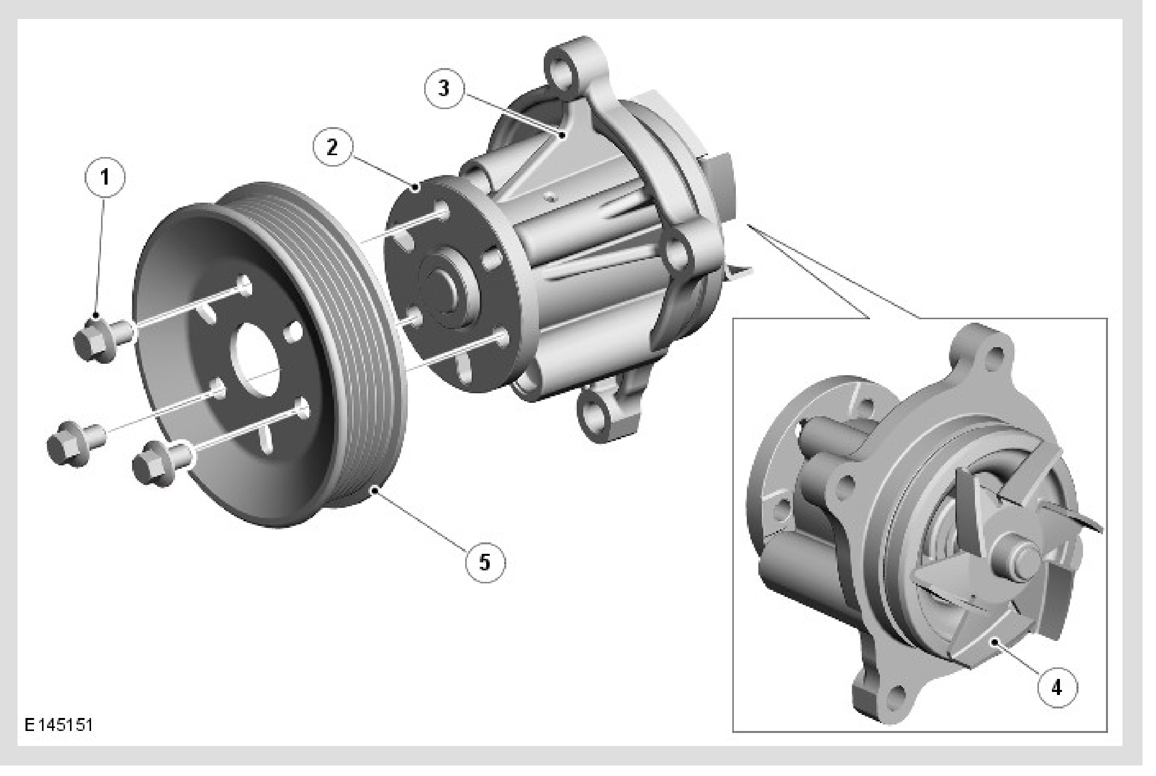
AUXILIARY FUEL COOLER



ITEM	DESCRIPTION
1	Left support bracket
2	Coolant outlet connection
3	Coolant intake connection
4	Right support bracket

The auxiliary fuel cooler is attached to the front of the A/C condenser. The end tanks of the cooler incorporate a coolant intake connection from the radiator upper hose and a coolant outlet connection to the fuel cooler in the fuel tank and lines system.

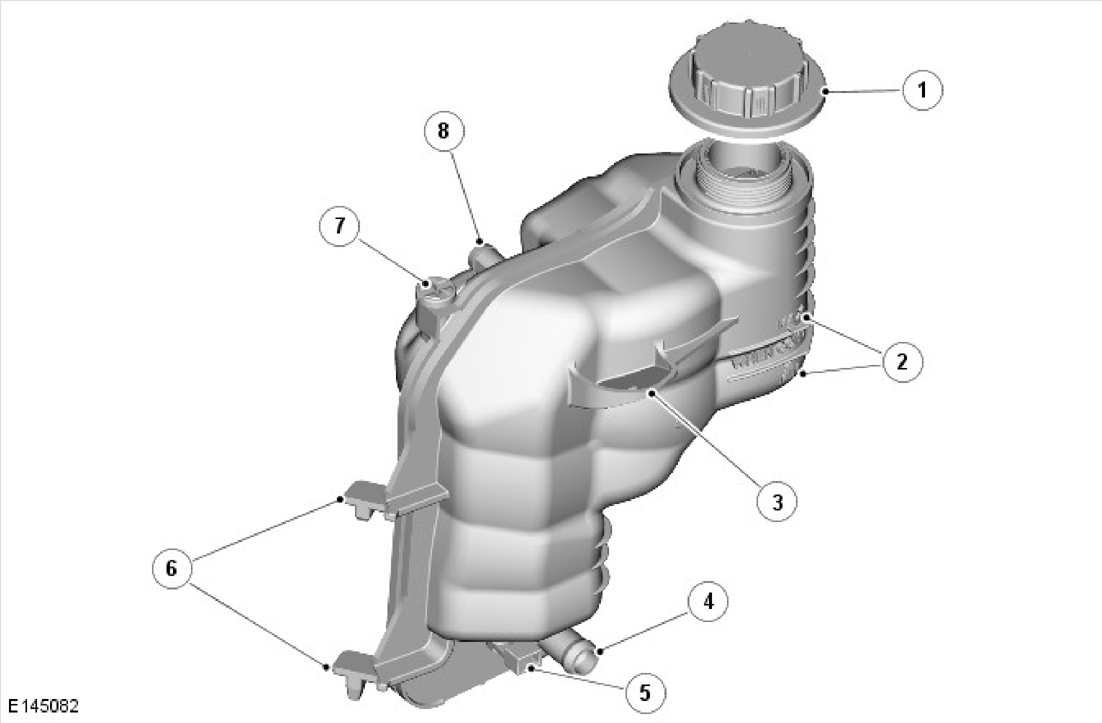
COOLANT PUMP



ITEM	DESCRIPTION
1	Bolt (3 off)
2	Drive hub
3	Housing
4	Impeller
5	Pulley

The coolant pump has a housing that supports a shaft with an impeller attached to one end and a drive hub at the other. The housing is attached to the front of the cylinder block with the impeller located in a pumping chamber. The pump is driven by a pulley attached to the drive hub and driven by the accessory drive belt.
For additional information, refer to: Accessory Drive (303-05D Accessory Drive - TDV8 4.4L Diesel, Description and Operation).

EXPANSION TANK



ITEM	DESCRIPTION
1	Filler cap
2	MAX and MIN level markings
3	Mounting lug
4	Supply hose connection
5	Coolant level sensor
6	Mounting lugs
7	Bleed screw
8	Vent hose connection

A pressurized expansion tank system is used which continuously separates the air from the cooling system and replenishes the system through a hose connected between the expansion tank and the thermostat. A continuous vent into the expansion tank, through a hose connected to the radiator, prevents air locks from forming in the cooling system.

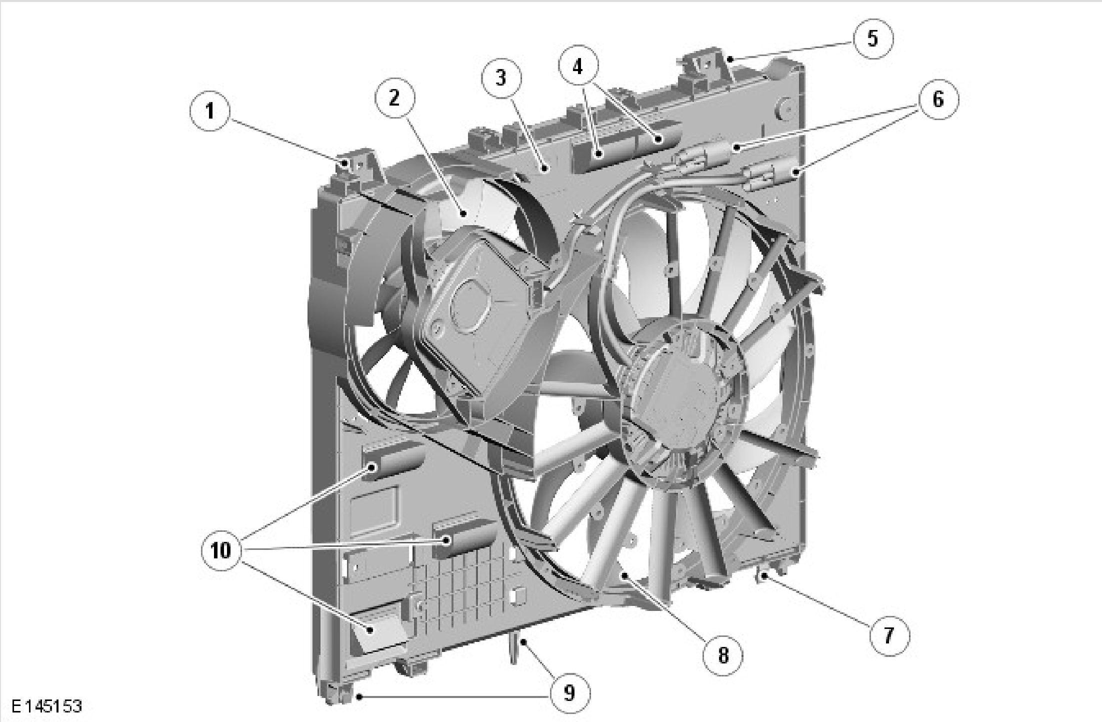
The expansion tank is attached to the front end carrier in the front left corner of the engine compartment. A filler cap, bleed screw and level sensor are incorporated into the expansion tank. MAX and MIN level markings are molded into the exterior of the tank.

The expansion tank provides the following functions:

- Service fill
- Coolant expansion during warm-up
- Air separation during operation
- System pressurization by the filler cap

The expansion tank has an air space of approximately 0.5 to 1 liter (1.06 to 2,11 US pints), above the MAX level, to allow for coolant expansion.

ENGINE COOLING FANS



ITEM	DESCRIPTION
1	Mounting lug
2	Cooling fan 2
3	Shroud
4	Speed flaps
5	Mounting lug
6	Electrical connectors

7	Mounting lug
8	Cooling fan 1
9	Mounting lugs
10	Speed flaps

Two variable speed electric engine cooling fans are installed to help regulate the coolant temperature. The fan shroud is attached to the rear of the radiator. The shroud incorporates speed flaps, which pivot open to allow engine compartment cooling at speed.

Each engine cooling fan is operated by a control module, integrated into their electric motor, which is controlled by the ECM . Electrical connectors at the right side of the fan shroud provide the interface between the cooling fan harnesses and the vehicle wiring.

The control module of each engine cooling fan is provided with:

- A battery power supply from the AJB (auxiliary junction box)
- An ignition signal from the ECM relay in the EJB (engine junction box)
- A PWM (pulse width modulation) signal from the ECM
- A ground.

To control each engine cooling fan, the ECM varies the duty cycle of the PWM signal. The fan control module translates the duty cycle into a fan target speed and sets the output current to the fan motor accordingly.

ENGINE COOLANT

The coolant is silicate free and must not be mixed with conventional engine coolant.

OPERATION

COOLANT CIRCUIT FLOW

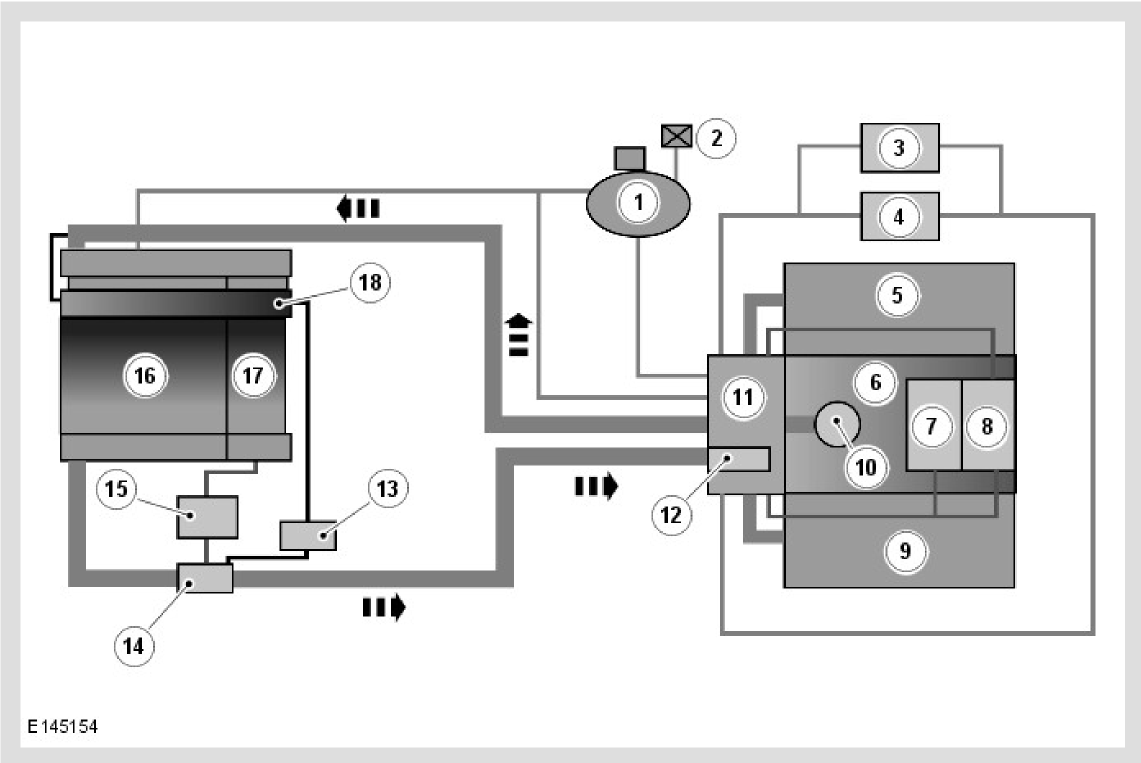
The coolant pump circulates coolant through the cylinder block and cylinder heads via a chamber located in the 'vee' of the engine. Having passed through the engine the coolant returns to the thermostat housing. Some of the coolant in the cylinder block is diverted through the engine oil cooler before returning to the thermostat housing.

From the thermostat housing, the majority of the coolant returns to the coolant pump, either directly or via the thermostat and radiator, depending on the temperature of the coolant and engine speed. The remainder of the coolant first flows through the EGR coolers and the heater circuit, then returns to the thermostat housing.

A separate hose from the radiator allows extra-cooled coolant from the radiator to flow through the transmission oil cooler. The coolant from the cooler returns to the system through a connector in the radiator lower hose.

A connection on the upper hose supplies coolant to the auxiliary fuel cooler. From the auxiliary fuel cooler, the coolant flows through the fuel cooler and returns to the system through the connector in the radiator lower hose.

Schematic Flow Diagram



ITEM	DESCRIPTION
1	Expansion tank
2	Static bleed point
3	Auxiliary heater (where fitted)
4	Heater
5	Cylinder head
6	Cylinder block
7	Engine oil cooler
8	Exhaust gas recirculation cooler
9	Cylinder head
10	Engine coolant pump

11	Thermostat housing
12	Thermostat
13	Fuel cooler
14	Connector
15	Transmission oil cooler
16	Radiator
17	Transmission oil sub-cooler
18	Auxiliary fuel cooler

THERMOSTAT

The thermostat is closed at temperatures below approximately 88°C (190°F). When the coolant temperature reaches approximately 88°C (190°F) the thermostat starts to open and is fully open at approximately 96°C (204°F). In this condition the full flow of coolant is directed through the radiator.

COOLANT LEVEL SENSOR

If the coolant level in the expansion tank decreases below a predetermined value, the coolant level sensor connects a ground to the CJB (central junction box) , which sends a message to the instrument cluster on the high speed CAN (controller area network) powertrain bus to display a warning message.
For additional information, refer to: Instrument Cluster (413-01 Instrument Cluster, Description and Operation).

COOLING FANS

The ECM determines when to operate the cooling fans from the ECT sensor input and a cooling fan request from the ATC (automatic temperature control) module. The ECM also adjusts the fan speed to compensate for the ram effect of vehicle speed.

The ECM varies the duty cycle of the PWM signal to the fan control module between 0 and 100% to operate the fan motor in one of four modes: Off, minimum speed (750 rev /min), linear variable speed between minimum and maximum speeds, and maximum speed (2820 rev/min). Under hot operating conditions, the fan may continue to operate for up to five minutes after the engine has been switched off.

The fan control module monitors for over and under input voltage, a stalled motor and a partially stalled motor. If it detects any of these faults the fan control module temporarily pulls the PWM signal to ground to notify the fault to the ECM . The length of time the PWM signal is pulled to ground varies between 2 and 8 seconds, depending on the fault. The fan control module repeats the notification process at 5 second intervals until the fault is cleared. If there is more than one fault, only the highest priority fault is notified to the ECM . Fault priorities, in descending order, are over voltage, under voltage, stalled motor and partially stalled motor. The ECM records a related DTC (diagnostic trouble code) for faults notified by the fan control module and signals the instrument cluster on the high speed CAN powertrain bus to display a warning message.

The nominal operating voltage for the engine cooling fans is 9 to 16 volts. If the input voltage is outside of these limits, the fan control module stops operation of the cooling fan and notifies the ECM of the fault. After an under or over voltage occurs, cooling fan operation is resumed when the input voltage increases to 9.5 volts, or decreases to 15.5 volts, respectively.

To check for a stalled motor, each time it supplies power to start the fan motor the fan control module checks the fan motor speed after 2 seconds. If the speed is zero, the fan control module determines the fan motor is stalled and tries a restart. For a restart, the fan control motor disconnects the power from the fan motor then immediately reconnects it at an increased current level. The fan control module will try up to six restarts of a stalled motor, increasing the output current each time. If the fan motor remains stalled after the sixth restart, the fan control module waits 40 seconds then notifies the ECM of the fault and begins the start process again.

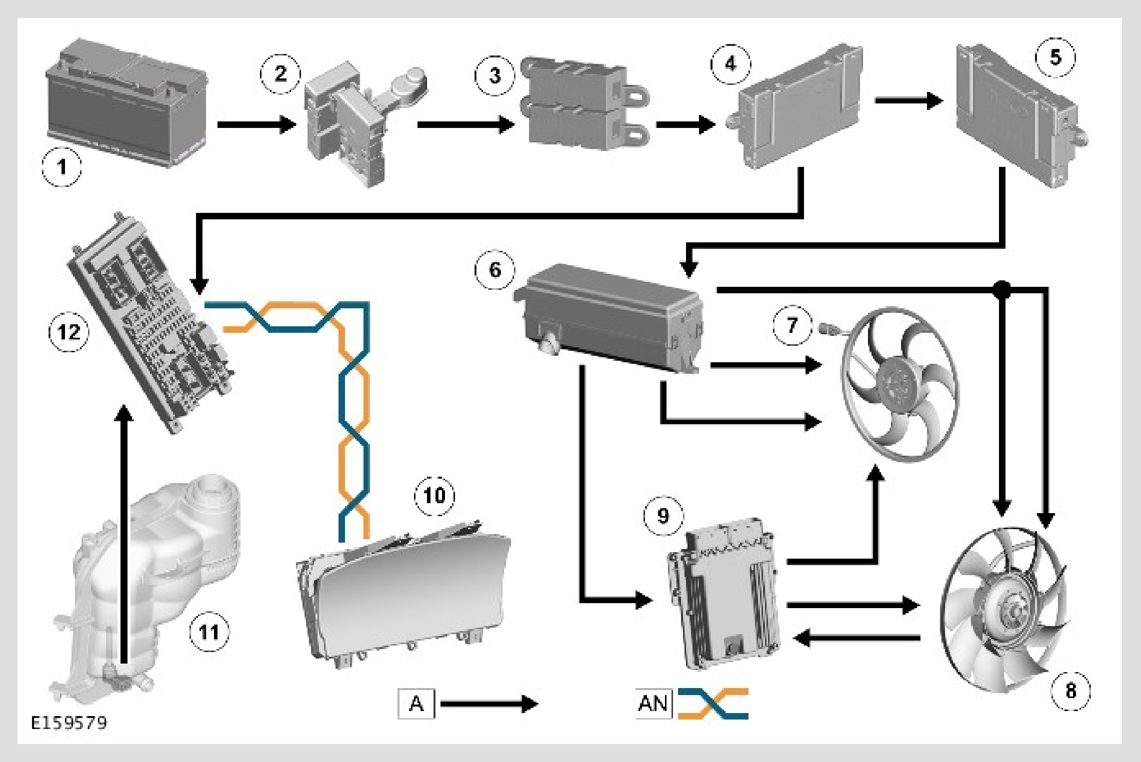
To check for a partially stalled motor, while the fan is running the fan control module monitors the current draw against fan speed. If the current draw is over the limit for more than 10 seconds, the fan control module disconnects the power from the fan motor and notifies the ECM of the fault. After 40 seconds, the fan control module starts the motor again.

The fan control module incorporates a temperature sensor to prevent damage caused by excessive heat at high ambient temperatures. Operation of the fan control module is discontinued if the temperature reaches 135 °C (275 °F). Operation resumes when the temperature decreases to 120 °C (248 °F).

If a fault occurs with the ignition signal or the PWM signal, the fan control module adopts the following fan speeds:

SIGNAL	FAULT	FAN SPEED	
		IGNITION ON	IGNITION OFF
PWM	Duty cycle implausible	Maximum	Off
PWM	Frequency out of range	Maximum	Off
PWM	Open circuit	Maximum	Off
PWM	Short circuit to battery	Maximum (provided cooling fan is not damaged)	Off
PWM	Short circuit to ground	Maximum (provided cooling fan is not damaged)	Off
Ignition	Open circuit	Off	Off
Ignition	Short circuit to battery	According to PWM duty cycle	Maximum
Ignition	Short circuit to ground	Off (ECM relay fuse may blow)	Off

CONTROL DIAGRAM



A = HARDWIRED; AN = HIGH SPEED CAN POWERTRAIN BUS.

ITEM	DESCRIPTION
1	Battery
2	Megafuse
3	BJB2 (battery junction box 2)
4	BJB (battery junction box)
5	AJB (auxiliary junction box)
6	EJB (engine junction box)
7	Engine Cooling Fan 1
8	Engine Cooling Fan 2
9	ECM (engine control module)

10	IC (instrument cluster)
11	Coolant Level Sensor
12	CJB (central junction box)